

H524 Introduction to Biostatistics

Week 10: More on Simple Linear Regression & Project Presentations

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Outline

- 1 Review from Week 9
- 2 Regression Assumptions
- 3 Diagnostic Plots
- 4 Outliers and Influential Points
- 5 Transformations
- 6 Hypothesis Testing and Confidence Intervals
- 7 Presenting Statistical Results
- 8 Case Study: Complete Analysis
- 9 AI for Statistical Communication
- 10 Final Project Guidance

Quick Review: Simple Linear Regression

Last week we covered:

- Simple linear regression model: $Y = \beta_0 + \beta_1 X + \epsilon$
- Least squares estimation: minimizing residuals
- Coefficient interpretation
- Correlation vs. regression
- R-squared and model fit

This Week: Model diagnostics, assumptions, transformations, and presenting results

Reading: Pagano & Gauvreau Chapter 18 (continued)

The Fitted Model Recap

Simple Linear Regression:

$$\hat{y} = b_0 + b_1x$$

where:

- $b_1 = r \cdot \frac{s_y}{s_x}$ (slope)
- $b_0 = \bar{y} - b_1\bar{x}$ (intercept)
- r = correlation coefficient

Key Concepts:

- **Residual:** $e_i = y_i - \hat{y}_i$ (observed - predicted)
- **Sum of Squared Errors:** $SSE = \sum e_i^2$
- **R-squared:** $R^2 = r^2$ (proportion of variance explained)

The Four Key Assumptions

For valid inference from linear regression:

- ➊ **Linearity:** The relationship between X and Y is linear
 - Check: Scatterplot, residual plot
- ➋ **Independence:** Observations are independent
 - Check: Study design, temporal patterns
- ➌ **Homoscedasticity:** Constant variance of residuals
 - Check: Residual vs fitted plot
- ➍ **Normality:** Residuals are approximately normal
 - Check: Q-Q plot, histogram of residuals

Acronym: LINE (Linearity, Independence, Normality, Equal variance)

Why Assumptions Matter

Consequences of violated assumptions:

Linearity violation:

- Biased estimates
- Poor predictions
- Misleading relationships

Independence violation:

- Underestimated standard errors
- Invalid p-values and confidence intervals
- False sense of precision

Homoscedasticity violation:

- Inefficient estimates
- Invalid hypothesis tests
- Incorrect prediction intervals

Normality violation:

Residual vs Fitted Plot

Purpose: Check linearity and homoscedasticity

What to look for:

- **Good:** Random scatter around horizontal line at zero
- **Bad:** Patterns, curves, funneling

Common patterns:

- **Curved pattern:** Non-linear relationship (try transformation)
- **Funnel shape:** Non-constant variance (heteroscedasticity)
- **Clusters/groups:** Missing categorical variable
- **Outliers:** Points far from the cloud

R code:

```
plot(fitted(model), residuals(model))  
abline(h=0, lty=2)
```

Q-Q Plot (Quantile-Quantile)

Purpose: Check normality of residuals

What to look for:

- **Good:** Points fall approximately on a straight line
- **Bad:** Systematic departures from the line

Common patterns:

- **S-shaped curve:** Heavy tails (outliers)
- **Curve below then above:** Right skewness
- **Curve above then below:** Left skewness
- **Points at ends deviate:** Extreme values

R code:

```
qqnorm(residuals(model))  
qqline(residuals(model))
```

Note: Minor departures are okay, especially with small samples

Scale-Location Plot

Purpose: Check homoscedasticity (constant variance)

What it shows: Square root of standardized residuals vs fitted values

What to look for:

- **Good:** Random scatter with horizontal trend line
- **Bad:** Upward or downward trend

Interpretation:

- Horizontal line $\Rightarrow$ constant variance
- Upward slope $\Rightarrow$ variance increases with fitted values
- Downward slope $\Rightarrow$ variance decreases with fitted values

R code:

```
plot(model, which=3)
```

Residuals vs Leverage Plot

Purpose: Identify influential observations

Key Concepts:

- **Leverage:** How far x_i is from $\bar{x}$ (potential influence)
- **Cook's Distance:** Actual influence on fitted values
- **Influential point:** High leverage AND large residual

What to look for:

- Points outside Cook's distance contours (dashed lines)
- Cook's distance > 0.5 or > 1 suggests influential observation

R code:

```
plot(model, which=5)
```

Action: Investigate influential points, but don't automatically remove them

Creating All Diagnostic Plots in R

Method 1: All four plots at once

```
par(mfrow=c(2,2))  
plot(model)  
par(mfrow=c(1,1))
```

Method 2: Individual plots

```
# 1. Residuals vs Fitted
```

```
plot(model, which=1)
```

```
# 2. Q-Q plot
```

```
plot(model, which=2)
```

```
# 3. Scale-Location
```

```
plot(model, which=3)
```

```
# 4. Residuals vs Leverage
```

```
plot(model, which=5)
```

Types of Unusual Observations

Three types to distinguish:

1. Outlier:

- Large residual (far from regression line)
- May or may not be influential
- Check for data entry errors

2. High leverage point:

- Extreme x value (far from $\bar{x}$)
- Potential to be influential
- May not have large residual

3. Influential point:

- High leverage AND large residual
- Removing it substantially changes regression line
- Check Cook's distance

DO NOT automatically remove outliers!

Steps to take:

1 Check for errors:

- Data entry mistakes
- Measurement errors
- Wrong units

2 Investigate context:

- Is it a legitimate observation?
- Does it represent a special subgroup?
- Was there a protocol violation?

3 Perform sensitivity analysis:

- Fit model with and without the point
- Report both analyses
- Discuss impact on conclusions

4 Consider robust methods:

- Robust regression (less sensitive to outliers)
- Transformations

Example: Identifying Influential Points in R

```
# Fit model
model <- lm(y ~ x, data=mydata)

# Calculate Cook's distance
cooks <- cooks.distance(model)

# Identify influential points (Cook's D > 0.5)
influential <- which(cooks > 0.5)
print(influential)

# Show influential observations
mydata[influential, ]

# Compare models with/without influential points
model_full <- lm(y ~ x, data=mydata)
model_reduced <- lm(y ~ x, data=mydata[-influential,])

summary(model_full)
summary(model_reduced)
```

When to Transform Data

Transformations can help when:

- Relationship is non-linear
- Variance is non-constant
- Residuals are non-normal
- Interpretation requires it (e.g., multiplicative effects)

Common transformations:

- **Log:** $\log(Y)$ or $\log(X)$ – for right skewness, exponential growth
- **Square root:** $\sqrt{Y}$ – for count data, mild right skewness
- **Reciprocal:** $1/Y$ or $1/X$ – for rates, strong right skewness
- **Square:** Y^2 or X^2 – for quadratic relationships

Box-Cox transformation: Data-driven choice of power transformation

Most common transformation in biostatistics

Model forms:

- ➊ **Log-linear:** $\log(Y) = \beta_0 + \beta_1 X$
 - Interpretation: 1-unit increase in $X \Rightarrow (e^{\beta_1} - 1) \times 100\%$ change in Y
- ➋ **Linear-log:** $Y = \beta_0 + \beta_1 \log(X)$
 - Interpretation: 1% increase in $X \Rightarrow \beta_1/100$ unit change in Y
- ➌ **Log-log:** $\log(Y) = \beta_0 + \beta_1 \log(X)$
 - Interpretation: β_1 is the elasticity (1% change in $X \Rightarrow \beta_1\%$ change in Y)

When to use:

- Right-skewed data
- Multiplicative relationships
- Relative changes more meaningful than absolute

Example: Log Transformation in R

Scenario: BMI vs medical cost (right-skewed)

Original model (may violate assumptions)

```
model_original <- lm(cost ~ bmi, data=health)
```

Log-transformed response

```
model_log <- lm(log(cost) ~ bmi, data=health)
```

Compare diagnostics

```
par(mfrow=c(2,2))
```

```
plot(model_original)
```

```
plot(model_log)
```

```
par(mfrow=c(1,1))
```

Interpretation

```
summary(model_log)
```

If $\beta_1 = 0.05$, then 1-unit increase in BMI

increases cost by $(\exp(0.05)-1)*100 = 5.1\%$

Polynomial Regression

For non-linear relationships with turning points

Quadratic model:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon$$

When to use:

- U-shaped or inverted U-shaped relationship
- Single peak or valley
- Example: age vs bone density (increases then decreases)

Interpretation:

- β_1 : slope at $X = 0$
- β_2 : rate of change in slope
- If $\beta_2 > 0$: U-shaped (convex)
- If $\beta_2 < 0$: inverted U-shaped (concave)

Warning: Don't extrapolate beyond data range!

Polynomial Regression in R

```
# Create quadratic term
mydata$age_squared <- mydata$age^2

# Fit quadratic model
model_quad <- lm(bone_density ~ age + age_squared,
                 data=mydata)

# Alternative: use poly() function
model_quad2 <- lm(bone_density ~ poly(age, 2, raw=TRUE),
                 data=mydata)

summary(model_quad)

# Find peak (maximum bone density)
# Peak at: -beta1 / (2 * beta2)
beta1 <- coef(model_quad)[2]
beta2 <- coef(model_quad)[3]
peak_age <- -beta1 / (2 * beta2)
print(paste("Peak at age:", round(peak_age, 1)))
```

Testing the Slope

Null hypothesis: $H_0 : \beta_1 = 0$ (no linear relationship)

Alternative: $H_A : \beta_1 \neq 0$ (linear relationship exists)

Test statistic:

$$t = \frac{b_1 - 0}{SE(b_1)} \sim t_{n-2}$$

where $SE(b_1) = \frac{s_e}{\sqrt{\sum(x_i - \bar{x})^2}}$ and $s_e = \sqrt{\frac{SSE}{n-2}}$

Decision:

- Reject H_0 if $|t| > t_{\alpha/2, n-2}$ or $p < \alpha$
- Fail to reject if $|t| \leq t_{\alpha/2, n-2}$ or $p \geq \alpha$

Note: This tests whether X is useful for predicting Y

Confidence Interval for Slope

95% CI for β_1 :

$$b_1 \pm t_{0.975, n-2} \times SE(b_1)$$

Interpretation:

- We are 95% confident the true slope is in this interval
- If interval excludes 0, reject $H_0 : \beta_1 = 0$
- Width reflects precision of estimate

Example:

- $b_1 = 2.5$ (estimated slope)
- $SE(b_1) = 0.8$
- $n = 30$, so $t_{0.975, 28} = 2.048$
- 95% CI: $2.5 \pm 2.048(0.8) = (0.86, 4.14)$
- Interpretation: For each 1-unit increase in X , Y increases between 0.86 and 4.14 units

Confidence vs Prediction Intervals

Two types of intervals for predictions:

1. Confidence Interval for Mean Response:

- Estimates the **average** Y for a given $X = x_0$
- Answers: What is the mean outcome for this X ?
- Narrower interval
- Example: Mean blood pressure for all 50-year-olds

2. Prediction Interval for Individual Response:

- Predicts a **single** Y for a given $X = x_0$
- Answers: What will this individual's outcome be?
- Wider interval (includes individual variability)
- Example: Blood pressure for one specific 50-year-old

Key difference: Prediction interval accounts for both:

- 1 Uncertainty in estimating the mean (same as CI)
- 2 Variability of individuals around the mean (extra)

Computing Intervals in R

```
# Fit model
model <- lm(sbp ~ age, data=health)

# New data point
new_data <- data.frame(age = 50)

# Confidence interval for mean SBP at age 50
predict(model, newdata=new_data,
        interval="confidence", level=0.95)
# Example output: fit=130, lwr=128, upr=132

# Prediction interval for individual at age 50
predict(model, newdata=new_data,
        interval="prediction", level=0.95)
# Example output: fit=130, lwr=110, upr=150

# Note: Prediction interval is wider!
```

Elements of a Good Statistical Presentation

For your final projects, include:

1 Clear research question

- What are you trying to learn?

2 Descriptive statistics

- Sample size, means, SDs
- Summary tables

3 Visualizations

- Scatterplots, histograms, boxplots
- Clear labels and legends

4 Statistical methods

- Which tests/models did you use?
- Why did you choose them?

5 Results

- Effect sizes, confidence intervals, p-values
- Interpretation in context

6 Limitations and assumptions

- What could go wrong?

Reporting Regression Results

Essential components:

- **Sample size:** $n = 250$ observations
- **Model equation:** $\hat{Y} = b_0 + b_1X$
- **Coefficient estimates:**
 - Point estimates
 - Standard errors
 - 95% confidence intervals
- **Statistical significance:** t-statistics, p-values
- **Model fit:** R^2 , residual standard error
- **Interpretation in context:**
 - What does the slope mean practically?
 - Clinical/scientific significance vs statistical
- **Diagnostics:** Did assumptions hold?

Example: “For each 1-unit increase in BMI, systolic BP increased by 1.2 mmHg (95% CI: 0.8-1.6, $p < 0.001$). The model explained 23% of the variance in SBP ($R^2 = 0.23$).”

Common Mistakes to Avoid

Don't:

- Confuse correlation with causation
 - “X is **associated with** Y” not “X **causes** Y”
- Over-interpret R-squared
 - Low R^2 doesn't mean the relationship isn't important
- Extrapolate beyond the data range
 - Don't predict for X values outside your sample
- Ignore influential points without investigation
- Report only p-values without effect sizes
 - **Always** report confidence intervals!
- Forget to check assumptions
 - Always create diagnostic plots
- Use jargon without explanation
 - Explain terms for non-statistical audience

Creating Effective Visualizations

Best practices:

- **Always label axes** with units
- **Include informative titles**
- **Use appropriate plot types**
 - Scatterplot for continuous X and Y
 - Boxplot for categorical X, continuous Y
- **Show the data and the model**
 - Points + regression line
 - Confidence/prediction bands
- **Use color wisely**
 - Colorblind-friendly palettes
 - Sufficient contrast
- **Keep it simple**
 - One main message per plot
 - Remove unnecessary elements

Example: Publication-Quality Plot

```
# Create professional scatterplot with regression
model <- lm(sbp ~ age, data=health)

# Set up plot
plot(health$age, health$sbp,
     xlab="Age (years)",
     ylab="Systolic Blood Pressure (mmHg)",
     main="Relationship between Age and Blood Pressure",
     pch=16, col=rgb(0,0,0,0.3), cex=1.2)

# Add regression line
abline(model, col="red", lwd=2)

# Add confidence band
new_age <- seq(min(health$age), max(health$age), 0.1)
pred_ci <- predict(model,
                   newdata=data.frame(age=new_age),
                   interval="confidence")
lines(new_age, pred_ci[, "lwr"], col="blue", lty=2)
```

Case Study: FEV and Age

Research Question: How does lung function (FEV) change with age in children?

Data: 654 children aged 3-19 years

- **Outcome:** FEV (forced expiratory volume, liters)
- **Predictor:** Age (years)

Hypothesis: FEV increases with age as lungs develop

Analysis Plan:

- 1 Descriptive statistics and visualization
- 2 Fit simple linear regression
- 3 Check assumptions
- 4 Interpret results
- 5 Consider alternatives if needed

Step 1: Descriptive Statistics

```
# Summary statistics
```

```
summary(fev_data$age)
```

```
#   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#   3.00   8.00   10.00   9.93  12.00   19.00
```

```
summary(fev_data$fev)
```

```
#   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#   0.791   1.981   2.547   2.637   3.119   5.793
```

```
sd(fev_data$age)      # 2.95 years
```

```
sd(fev_data$fev)      # 0.87 liters
```

```
# Correlation
```

```
cor(fev_data$age, fev_data$fev)
```

```
# r = 0.877 (strong positive correlation)
```

Step 2: Initial Visualization

```
# Scatterplot
plot(fev_data$age, fev_data$fev,
     xlab="Age (years)",
     ylab="FEV (liters)",
     main="FEV vs Age in Children",
     pch=16, col=rgb(0,0,1,0.3))
```

Observations:

- # - Strong positive relationship
- # - Possible non-linearity (steeper for younger)
- # - Greater variability for older children

Initial impression: Linear model may work, but check diagnostics

Step 3: Fit Linear Model

```
# Fit simple linear regression
model <- lm(fev ~ age, data=fev_data)
summary(model)

# Coefficients:
#               Estimate Std. Error t value Pr(>|t|)
# (Intercept)  0.43161    0.06328   6.821  1.8e-11 ***
# age          0.22204    0.00608  36.515  < 2e-16 ***
# ---
# Residual standard error: 0.358 on 652 degrees of freedom
# Multiple R-squared:  0.6714,    Adjusted R-squared:  0.6709
# F-statistic: 1333 on 1 and 652 DF,  p-value: < 2.2e-16

# Interpretation:
# FEV increases by 0.222 liters per year of age
# Model explains 67.1% of variance
```


Step 4: Check Assumptions

```
# Diagnostic plots
par(mfrow=c(2,2))
plot(model)
par(mfrow=c(1,1))

# Observations:
# 1. Residuals vs Fitted: slight curve (non-linearity?)
# 2. Q-Q plot: reasonably linear (normality OK)
# 3. Scale-Location: some funneling (heteroscedasticity?)
# 4. Residuals vs Leverage: a few high leverage points

# Potential issues:
# - Non-linearity (could try quadratic or log)
# - Non-constant variance (more spread for older ages)
```

Step 5: Consider Alternatives

Given diagnostic concerns, try quadratic model:

```
# Quadratic model
model_quad <- lm(fev ~ age + I(age^2), data=fev_data)
summary(model_quad)

# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)  1.04423    0.10055   10.39 < 2e-16 ***
# age         -0.01626    0.02273   -0.72  0.475
# I(age^2)     0.01201    0.00109   11.00 < 2e-16 ***
# ---
# Multiple R-squared:  0.7198,    Adjusted R-squared:  0.7189

# Better R-squared (72% vs 67%)
# Check diagnostics again...
```

Step 6: Interpret and Report

Final Model: Quadratic relationship

$$\text{FEV} = 1.04 - 0.016 \times \text{Age} + 0.012 \times \text{Age}^2$$

Key Findings:

- FEV increases with age in children, but non-linearly
- Rate of increase accelerates with age
- Model explains 72% of variance
- All coefficients highly significant ($p < 0.001$)

Conclusion:

- Strong positive relationship between age and lung function
- Quadratic model fits better than linear
- Results consistent with biological lung development
- Caution: Don't extrapolate to adults (age > 19)

AI for Presenting Statistical Results

How AI can help with your project presentations:

- **Clarifying explanations:** Translate technical results for non-experts
- **Visualization code:** Generate R code for publication-quality plots
- **Interpretation guidance:** Help explain what results mean in context
- **Checking logic:** Verify your statistical reasoning
- **Writing assistance:** Draft results sections
- **Q&A preparation:** Anticipate questions about your analysis

Example prompts:

- “Explain what R-squared means to someone without statistics training”
- “Create a scatterplot with regression line and 95% CI bands in R”
- “How do I interpret a slope of 2.5 with 95% CI (1.2, 3.8)?”
- “What are potential limitations of simple linear regression?”

AI Demo: Explaining Results

Prompt: “I found slope = 1.2 (SE = 0.3) for BMI predicting blood pressure. Explain this to a doctor.”

Claude AI Response:

This means that for each 1-point increase in BMI, we expect systolic blood pressure to increase by 1.2 mmHg on average.

The standard error (0.3) tells us the precision of this estimate. A 95% confidence interval would be: $1.2 \pm 1.96(0.3) = (0.6, 1.8)$ mmHg

Clinically, this suggests that a patient with BMI of 30 would have blood pressure about 12 mmHg higher than a patient with BMI of 20, all else equal.

This is a statistically significant and clinically meaningful association, though it doesn't prove that reducing BMI will lower blood pressure (correlation

AI Demo: Creating Visualizations

Prompt: “Create an R plot showing regression line with both confidence and prediction bands”

Claude AI Response:

```
# Fit model
model <- lm(y ~ x, data=mydata)

# Create sequence for predictions
x_seq <- seq(min(mydata$x), max(mydata$x), length=100)

# Confidence interval
pred_ci <- predict(model, newdata=data.frame(x=x_seq),
                    interval="confidence", level=0.95)

# Prediction interval
pred_pi <- predict(model, newdata=data.frame(x=x_seq),
                    interval="prediction", level=0.95)

# Plot
plot(mydata$x, mydata$y, pch=16, col="gray50",
      xlab="X", ylab="Y", main="Regression with Intervals")
lines(x_seq, pred_ci[, "fit"], col="blue", lwd=2)
lines(x_seq, pred_ci[, "lwr"], col="blue", lty=2)
lines(x_seq, pred_ci[, "upr"], col="blue", lty=2)
lines(x_seq, pred_pi[, "lwr"], col="red", lty=2)
lines(x_seq, pred_pi[, "upr"], col="red", lty=2)
legend("topleft", c("Data", "Fit", "95% CI", "95% PI"),
      col=c("gray50", "blue", "blue", "red"), lty=c(NA, 1, 2, 2),
      pch=c(16, NA, NA, NA))
```

AI Best Practices for Presentations

DO:

- Use AI to clarify complex concepts
- Ask for multiple visualization options
- Request interpretation help
- Generate example explanations
- Check your statistical reasoning
- Get help with R code

DON'T:

- Let AI write entire presentation
- Use AI explanations without understanding
- Trust AI for statistical decisions
- Copy AI text verbatim
- Skip checking AI's work
- Ignore your own judgment

Remember: AI is a tool to enhance YOUR work, not replace it

For final projects:

- You must understand everything you present
- Be prepared to answer questions independently
- Acknowledge AI assistance if used

AI Task: Preparing Your Presentation

Use AI to help with these tasks:

- ① **Simplify technical language**
 - Ask AI to rephrase statistical jargon
- ② **Create clear visualizations**
 - Get R code for professional plots
- ③ **Anticipate questions**
 - Ask “What questions might someone ask about this analysis?”
- ④ **Check interpretations**
 - Verify your understanding is correct
- ⑤ **Draft explanations**
 - Get help wording conclusions

Example: “Review my regression analysis and suggest improvements to the presentation”

Final Project Presentations

Format:

- 10-12 minute presentation
- 3-5 minutes for questions
- PowerPoint, Beamer, or Google Slides

Required content:

- 1 Introduction and research question
- 2 Data description and summary statistics
- 3 Statistical methods used
- 4 Results (tables, figures)
- 5 Interpretation and conclusions
- 6 Limitations

Grading criteria:

- Clarity and organization (20%)
- Appropriate statistical methods (30%)
- Correct interpretation (25%)
- Quality of visualizations (15%)

Tips for Successful Presentations

Do:

- Practice your presentation multiple times
- Speak clearly and at moderate pace
- Make eye contact with audience
- Explain statistical terms
- Focus on interpretation, not just methods
- Use clear, uncluttered slides
- Tell a story with your data

Don't:

- Read directly from slides
- Use too much text on slides
- Skip over assumptions or limitations
- Ignore questions you don't know – say “good question, I'll look into that”
- Go over time limit

Avoid these mistakes:

- ❶ **Too much technical detail**
 - Focus on interpretation, not formulas
- ❷ **No context for numbers**
 - Always interpret in practical terms
- ❸ **Cluttered visualizations**
 - One clear message per plot
- ❹ **Ignoring limitations**
 - Acknowledge what your analysis can't tell you
- ❺ **Poor time management**
 - Practice with a timer
- ❻ **Not checking assumptions**
 - Always show diagnostic plots

This week we covered:

- **Regression diagnostics:**
 - Four key assumptions (LINE)
 - Diagnostic plots
 - Identifying influential points
- **Transformations:**
 - Log, square root, polynomial
 - When and why to transform
- **Inference:**
 - Hypothesis tests for slope
 - Confidence and prediction intervals
- **Communication:**
 - Reporting results clearly
 - Creating effective visualizations
 - Using AI to enhance presentations

Final Project Presentations: Next week!

Prepare: Practice your presentation, check all analyses, create clear

Thank you!

Good luck with your final projects!

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